Aerone Carl P. Peñol

Electronics Engineering Student | Programming & Research

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Open to internship opportunities

PROFESSIONAL SUMMARY

Second-year Electronics Engineering student at TUP–Visayas, interested in programming, research, and how hardware and software work together. Builds practical projects, explores PCB design, and applies technical problem-solving through coursework, independent learning, and publication work.

TECHNICAL SKILLS

Frontend: HTML5, CSS3, JavaScript

Backend & database: Supabase, PostgreSQL

Development & deployment: GitHub, Vercel

AI development tools: ChatGPT, OpenAI Codex, Claude, Google Antigravity

Engineering & design: AutoCAD, SketchUp, EasyEDA

Other tools I explore: Basic Python, Basic Java

EDUCATION

Bachelor of Science in Electronics Engineering

Technological University of the Philippines–Visayas · Expected graduation: 2029

Senior High School Graduate

Isabela National High School · 2024 · With High Honors

PUBLICATION EXPERIENCE

Junior Layout Artist · The Philippine Artisan – Visayas 2025

TUP Visayas · Publication layout and visual communication

- Contributed to the January–October 2025 tabloid's layout, applying typography, spacing, and visual hierarchy.
- The publication earned COPRE recognition: Champion in Best Newspaper Layout, 3rd Place in Best News Page, and 5th Place in Best Filipino Page.

Layout Artist · The Glazier 2024

Isabela National High School · Page composition and readability

- Worked on page layouts as part of the publication team that earned 2nd Place at the Division Schools Press Conference (DSPC).

LEADERSHIP EXPERIENCE

Auditor · Supreme Student Government 2024

- Supported financial oversight and accountability within the student government organization.

Supreme Pupil Government

Vice President · 2017–2019 | Public Information Officer · 2016–2017

Peace Officer · 2015–2016

Assisted student governance, represented student interests, supported communications, and helped maintain a positive school environment.

Aerone Carl P. Peñol Projects & Research

SELECTED PROJECTS

Regulated Full-Wave Bridge 5V Power Supply · Coursework

Designed a first PCB layout for Computer-Aided Design. The power supply converts AC input into regulated 5V DC through rectification, filtering, and voltage regulation.

EasyEDA · PCB Design · 5V Power Supply

Axiom Studia · Personal project · In progress

Designed and refined a web-based academic productivity platform for organizing subjects, learning resources, daily tasks, deadlines, and study activities. Used iterative problem-solving and AI-assisted coding to improve interface layouts, navigation, and usability.

Web Development · UI/UX Design · AI-Assisted Coding

Smart Solar-Powered Sensory Anti-Pest Device · Research project

Explored an engineering-based approach to pest management for residential and office environments, combining solar power and sensor-based detection. Applied research and technical concepts to a practical sustainability problem.

Solar Energy · Sensors · Sustainable Engineering

RESEARCH EXPERIENCE

Selected High School Research Projects

- Foliar Application of Fermented Liquid Fertilizer Formulated from Eggshell Powder and Banana Peels (Musa sapientum) on Eggplants (Solanum melongena).
- The Potency of Cassava (Manihot esculenta) Starch and Taro Root (Colocasia esculenta) Starch as Materials for Biodegradable, Non-Toxic Plastic Bags.
- Effects of Social Factors on the Academic Achievement of Senior High School Students.

Research interests include biodegradable materials, social factors in education, renewable energy, and practical engineering applications.

ACADEMIC HONORS & AWARDS

- With High Honors · Senior High School Graduation · 2024
- Class Salutatorian · Elementary School Graduation · 2018
- Class Rank No. 2 · 2012–2018

ADDITIONAL CAPABILITIES

UI/UX and layout design · Visual communication · Critical thinking · Logical reasoning · Research and analysis · Organization · Time management · Adaptability · Teamwork · Independent learning